The hglm algorithm

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HGLM course at the Roslin Institute

A linear model

To start with, consider a linear model with only fixed effects:

$$\mathbf{y} \sim \mathrm{N}(\mathbf{X}\beta, \sigma_e^2)$$

Can also be written as

$$\mathbf{y} = \mathbf{X}\beta + \mathbf{e}$$

$$\mathbf{e} \sim \mathrm{N}(0, \sigma_e^2)$$

How can this model be fitted?

- Maximum likelihood: $\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$, $\hat{\sigma}_e^2 = \frac{1}{n}(\mathbf{y} - \mathbf{X}\hat{\beta})'(\mathbf{y} - \mathbf{X}\hat{\beta})$
- Unbiased residual variance estimate: $\hat{\sigma}_e^2 = \frac{1}{n-p}(\mathbf{y} - \mathbf{X}\hat{\beta})'(\mathbf{y} - \mathbf{X}\hat{\beta})$

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Using GLM

- Basic idea
  - If the fixed effects in the mean part of the model (ie β) were known,
    - then the squared residuals are $e_i^2 \sim \sigma_e^2 \cdot \chi_1^2$ (for observation i), i.e. gamma distributed.
  - So, the squared residuals may be fitted using a GLM having a gamma distribution with a log link function.
- But β is estimated and not known
  - $V(\hat{e}_i) = (1 - h_{ii})\sigma_e^2$ where h_{ii} are the diagonal elements of the "hat matrix" $\mathbf{H}$
  - $\mathbf{H}$ is defined so that $\hat{\mathbf{y}} = \mathbf{Hy}$ (ie $\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$, giving $\mathbf{Hy} = \mathbf{X}\hat{\beta}$)
  - $\frac{(\hat{e}_i)^2}{1-h_{ii}}$ fitted using a gamma GLM with log link (and weights $\frac{1-h_{ii}}{2}$).

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A heteroscedastic linear model

Consider now a linear model with only fixed effects both in the mean and dispersion parts:

$$\mathbf{y} \sim \mathrm{N}(\mathbf{X}\beta, \quad exp(\mathbf{X}_d \beta_d))$$

Can also be written as

$$\mathbf{y} = \mathbf{X}\beta + \mathbf{e}$$

$$\mathbf{e} \sim \mathrm{N}(0, \sigma_e^2)$$

$$log(\sigma_e^2) = \mathbf{X}_d \beta_d$$

How can this model be fitted?

Two alternatives giving the same results

- Specify the likelihood and maximize it using a numerical method
- Iterate between a linear model and a GLM
  - Estimate β for a given residual variance
  - Estimate β_d : Fit $\frac{(\hat{e}_i)^2}{1-h_{ii}}$ as response variable in a gamma GLM (log link) with linear predictor $\mathbf{X}_d \beta_d$

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Can this be used for linear mixed models?

$$y = \mathbf{X}b + \mathbf{Z}u + e$$

$V = \mathbf{Z}\mathbf{Z}'\sigma_u^2 + \mathbf{I}_n\sigma_e^2$

Re-write it as an augmented weighted linear model!!!!

$$y_a = \mathbf{T}\delta + e_a$$

where $y_a = \begin{pmatrix} y \\ \mathbf{0} \end{pmatrix}$, $\mathbf{T} = \begin{pmatrix} \mathbf{X} & \mathbf{Z} \\ \mathbf{0} & \mathbf{I}_k \end{pmatrix}$, $\delta = \begin{pmatrix} b \\ u \end{pmatrix}$, $e_a = \begin{pmatrix} e \\ -u \end{pmatrix}$

The variance-covariance matrix of the augmented residual vector is given by

$V(e_a) \equiv W^{-1} = \begin{pmatrix} \mathbf{I}_n\sigma_e^2 & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_k\sigma_u^2 \end{pmatrix}$

Weighted least squares estimates

$$\mathbf{T}'W\mathbf{T}\hat{\delta} = \mathbf{T}^t W y_a$$

Identical to Henderson's mixed model equations!! Left hand side

$\mathbf{T}'W\mathbf{T} = \begin{pmatrix} \mathbf{X}'\mathbf{X}\frac{1}{\sigma_e^2} & \mathbf{X}'\mathbf{Z}\frac{1}{\sigma_e^2} \\ \mathbf{Z}'\mathbf{X}\frac{1}{\sigma_e^2} & \mathbf{Z}'\mathbf{Z}\frac{1}{\sigma_e^2} + \mathbf{I}_k\frac{1}{\sigma_u^2} \end{pmatrix}$

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Use same method as before

- σ_e^2 - use gamma GLM with response $\hat{e}_i^2/(1-h_{ii})$, weights $(1-h_{ii})/2$, $i=1,...,n$.
- σ_u^2 - use gamma GLM with response $\hat{u}_j^2/(1-h_{jj})$, weights $(1-h_{jj})/2$, $i=1,...,k$. h_{jj} last k hat values of the augmented model.
- $\mathbf{H}=\mathbf{T}(\mathbf{T}'W\mathbf{T})^{-1}\mathbf{T}'W$